



DEVOPS

2025

ABOUT US

Studybalancer is a premier provider of cutting-edge training programs in cloud computing and DevOps. We aim to provide training on the most demanding courses, including AWS (Amazon Web Services), Microsoft Azure, Google Cloud Platform (GCP), and DevOps. Our courses cater to both freshers and professionals looking to upskill and advance their careers.

Our course curriculum is unique and exceptional, offering content that is not available anywhere else in the market. We ensure that our training programs are comprehensive, up-to-date, and tailored to meet the specific needs of our students.

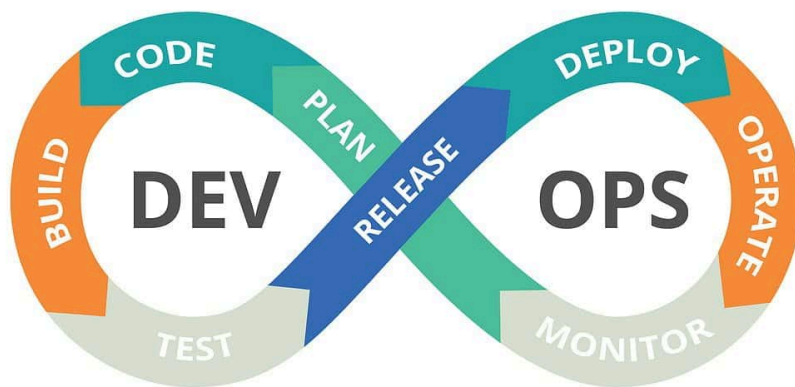
We bring the best learning experience for both individuals and organisations through our interactive, customized courses.



OVERVIEW

The DevOps Program by StudyBalancer is designed to equip aspiring DevOps professionals with the in-demand skills and tools required to streamline development and operations. This hands-on course covers everything from Linux basics to CI/CD pipelines and cloud-native tools like Kubernetes, Terraform, and Ansible.

Whether you're just starting out or want to transition into a DevOps role, this comprehensive program builds your expertise across system administration, cloud infrastructure, containerization, automation, monitoring, and continuous integration/delivery practices



PREREQUISITES

- Basic knowledge of operating systems and networking
- Familiarity with terminal/command line is helpful
- No prior DevOps or cloud experience required — we teach from scratch



BRINGING IDEAS TO LIFE AND
FOSTERING INNOVATION
THROUGH TECHNOLOGY.

WHY GET DEVOPS CERTIFIED?

- **High Demand, High Pay:** DevOps roles are among the top-paying in tech globally.
- **Career Flexibility:** Work as a DevOps Engineer, Site Reliability Engineer, Cloud Engineer, or Automation Specialist.
- **Boost Team Collaboration:** Learn how to break silos between development and operations for faster, reliable software delivery.
- **Cloud-Ready Skill Set:** Stay competitive in a cloud-native job market with AWS, Kubernetes, Terraform, and more.
- **Globally Recognized Certifications:** Gain credibility with certifications like AWS Certified Solutions Architect.



BRINGING IDEAS TO LIFE AND FOSTERING
INNOVATION THROUGH TECHNOLOGY.

OUR COMPREHENSIVE Learning Approach



LIVE INSTRUCTOR TRAINING

Experience interactive, real-time learning with expert instructors guiding you every step of the way. Get personalized attention and clarity through live sessions that bring the classroom to your screen.



ACCESS TO LMS

Enjoy 24/7 access to our Learning Management System, where you can revisit course materials anytime. Track your progress, download resources, and stay updated with ease



TECHNICAL SUPPORT

Our dedicated technical support team is always ready to assist you with any issues. From platform navigation to troubleshooting, we ensure a smooth learning experience.



EXPERIENCE INSTRUCTOR

Learn from seasoned professionals with years of industry experience and expertise. Our instructors are passionate about delivering high-quality, practical knowledge that you can apply immediately.

COURSE CURRICULUM

INTRO TO DEVOPS COURSE CURRICULUM (1WEEK)

1. SDLC Models Overview

- Introduction to Software Development Life Cycle (SDLC).
- Overview of popular SDLC models: Waterfall, Agile, Iterative, Spiral, and V-Model.
- Comparing traditional models with modern Agile/DevOps practices.

2. Why DevOps is Needed

- Challenges in traditional software delivery methods.
- How DevOps bridges the gap between development and operations teams.
- Benefits of DevOps: faster delivery, improved collaboration, and enhanced quality.

3. How DevOps is Changing the World

- Real-world examples of organizations adopting DevOps.
- Impact of DevOps on industry trends and innovation.
- The role of DevOps in digital transformation and cloud computing.

4. Job Roles After This Course

- Overview of DevOps-related career paths:
- DevOps Engineer
- Site Reliability Engineer (SRE)
- Automation Specialist
- Cloud Engineer
- Release Manager
- Skills employers look for in DevOps professionals.
- Certifications and additional learning opportunities for advancing in DevOps.

5. What is Automation

- Concept of automation in software development and deployment.
- Benefits of automation: scalability, efficiency, and error reduction.
- Areas in DevOps where automation is essential (e.g., CI/CD, testing, monitoring).

6. Tools and Approach for the Course

- Tools covered:
- Git for version control.
- Jenkins for continuous integration and deployment.
- Docker for containerization.
- Kubernetes for orchestration.

- Terraform for infrastructure as code.
- Monitoring tools like Prometheus and Grafana.
- **Approach:**
 - Hands-on learning with labs and simulations.
 - Real-world project integration for practical exposure.
 - Collaborative learning through assignments and group discussions.

7. Project Information

- Capstone project: Implementing a DevOps pipeline.
- Designing, building, and automating a CI/CD process.
- Integrating tools like Docker and Kubernetes in the project workflow.
- Deploying a web application with monitoring and scaling features.

8. Language Use

- Programming/scripting languages to focus on:
- Python and Bash scripting for automation tasks.
- YAML for configuration files.
- JavaScript (optional) for application-related implementations.
- Basic overview of writing scripts and code for DevOps processes.

LINUX FUNDAMENTALS & SYSTEM ADMINISTRATION (2WEEK)

1. Introduction to Linux

- What is Linux?
- Overview of Linux and its history.
- Understanding the Linux Kernel and distributions.
- Why choose Linux? Benefits and use cases.

2. Essentials Topics

- Understanding file systems and directory structure.
- Linux installation basics (different methods and tools).
- Permissions and ownership: Read, Write, and Execute.
- Basic processes: starting, stopping, and managing processes.

3. Users and Groups Management

- Creating and managing users.
- Understanding groups and assigning group permissions.
- File and folder ownership: chown and chmod commands.
- Switching users and root access.

4. Operations Deployment

- Basics of software installation: apt, yum, dnf, and package managers.
- Running services and daemons.
- Setting up and managing cron jobs for automation.
- Understanding service management with systemctl.

5. Storage and Networking

- Disk partitioning and file systems: ext4, NTFS, FAT.
- Mounting and unmounting storage devices.
- Disk usage monitoring using df, du, and lsblk.
- Backup and recovery basics.
- fdisk/parted, LVM, mount options
- ifconfig/ip, netstat/ss, resolv.conf, troubleshooting DNS

6. Log Management

- Introduction to logrotate and its purpose.
- Configuring log rotation policies.
- Scheduling and automation of log rotation.
- Hands-on: setting up logrotate for system logs.

7. Security Basics

- SELinux/AppArmor modes
- Firewall/Iptables basics

8. Process & Service Management

- Monitoring processes: ps, top/htop, nice/renice, kill/pkill
- systemd fundamentals: units (service, socket, timer, target), journalctl logging, cgroups
- Managing services: systemctl start|stop|restart|status, enabling at boot, masking, viewing dependencies

Hands-on Lab

Harden a freshly provisioned VM: create user accounts, lock down SSH, configure a basic firewall, and automate routine updates via cron/Ansible.

Assessment

Given a bare VM, deliver a secure, production-ready Linux host with documented steps and validation.

VERSION CONTROL WITH GIT & GITHUB (1WEEK)

- Introduction to Git concepts: repositories, commits, branches, merges
- Branching strategies: Git Flow, Github Flow
- Pull requests & code reviews
- Repo branch protection strategies and also rule set.
- Github Webhooks
- Intro to Github App

Hands-on Lab: Initialize a repo, simulate team workflow, resolve merge conflicts

Assessment: Pull-request exercise

NETWORKING FUNDAMENTALS AND SERVER MANAGEMENT(1WEEKS)

1. What is Networking

- Definition of networking.
- Importance of networking in modern communication.
- Overview of network infrastructures and applications.

2. OSI and TCP/IP Model

- Understanding the OSI Model: Layers, functionality, and examples.
- TCP/IP Model: Layers and comparison with OSI Model.
- Protocols and services associated with each layer.

3. Types of Networks and Topologies

- Types of networks: LAN, WAN, MAN, PAN, CAN.
- Network topologies: Star, Ring, Bus, Mesh, and Hybrid.
- Practical application of different types and topologies.

4. Devices

- Overview of networking devices.
- Hub: Functionality and use cases.
- Switches: Types, features, and VLANs.
- Routers:
 - Routing protocols: OSPF and BGP.
 - Static routing versus dynamic routing.
 - Virtual routing concepts.

5. IPv4 and IPv6 with Subnetting

- Difference between IPv4 and IPv6.
- Basics of IP addressing and binary representation.
- Subnetting concepts and CIDR notation.
- Practice subnetting exercises.

6. NAT (Network Address Translation)

- What is NAT and why it is important.
- Types of NAT: Static, Dynamic, and PAT (Port Address Translation).
- Real-world applications of NAT.

7. FTP and SFTP

- What are FTP and SFTP?
- Differences and use cases.
- Secure file transfer protocols and configurations.

8. SSH (Secure Shell)

- Basics of SSH and its importance for secure remote access.
- Generating and managing SSH keys.
- Practical applications like server access and file transfers.

9. NTP (Network Time Protocol)

- Importance of time synchronization in networks.
- How NTP works.
- Setting up and troubleshooting NTP services.

10. NGINX or Apache (Reverse Proxy)

- Introduction to reverse proxy servers.
- Comparing NGINX and Apache.
- Setting up reverse proxy configurations.
- Load balancing using a reverse proxy.

11. VPN and Firewalls

- Introduction to VPNs: Types (Remote Access, Site-to-Site, etc.).
- Firewall basics and types.
- Configuring the firewalld service for network security.
- Practical setup of firewall rules and policies.

12. Networking Commands in Linux

- Common Linux networking commands:
 - ping, netstat, ifconfig, ip, traceroute.
 - Managing network configurations (nmcli and nmtui).
 - Network troubleshooting commands: tcpdump, ss.

PROGRAMMING AND SCRIPTING COURSE CURRICULUM (2WEEKS)

1. Introduction to Python Language

- Basics of Python: Syntax, variables, data types, and control structures.
- Python modules and libraries.
- File handling and error/exception handling in Python.
- Introduction to object-oriented programming (OOP) in Python.
- Working with Yaml and json.

2. Shell Scripting

- Basics of shell scripting and Linux shells (e.g., Bash).
- Writing and executing shell scripts.
- Control structures: loops, conditionals, and functions in shell scripts.
- File operations and text processing (e.g., awk, sed, grep).
- Automating system tasks with shell scripts.

3. Why Python is Needed Over Shell Scripting

- Comparison of Python and shell scripting:
 - Ease of use, readability, scalability, and cross-platform support.
- Use cases where Python is preferable.
- Limitations of shell scripting and how Python overcomes them.

4. Combining Python and Shell Scripting

- Using Python to execute shell commands (subprocess module).
- Automating system tasks with a combination of Python and shell scripts.
- Real-world examples of integrating both tools for enhanced automation.

5. Linux Essentials:

- **set Command:**
 - Understanding and using the set command to modify shell behavior.
- **source Command:**
 - Explanation and practical use cases of sourcing scripts.
- **.bashrc File:**
 - What is .bashrc and its role in shell configuration.
 - Customizing the shell environment using .bashrc.
- **Shell Basics:**
 - Overview of shell environments and configuration management.

Assessment: Script to provision a simple web server

AWS CERTIFIED SOLUTIONS ARCHITECT ASSOCIATE (3WEEKS)

The AWS Certified Solutions Architect – Associate certification is focused on the design of cost-efficient, performance-optimized, and secure applications on the AWS platform. Here's a breakdown of the main domains and topics covered in the exam:

1. Design Secure Architectures (30%)

- Design secure access to AWS resources: Implement identity and access management (IAM), secure root accounts, and IAM policies.
- Design secure application tiers: Secure data at rest and in transit, implement least-privilege principles.
- Select appropriate data security options: Encryption and security controls.

2. Design Resilient Architectures (26%)

- Design high availability and/or fault-tolerant architectures: Understand disaster recovery (DR) strategies, replication, and redundancy.
- Determine resource allocation strategies: Use of Amazon EC2 autoscaling and integration with load balancers.
- Determine how to design decoupling mechanisms: AWS services for asynchronous decoupling, such as SNS, SQS, and Lambda.

3. Design High-Performing Architectures (24%)

- Identify elastic and scalable compute solutions for a workload: Amazon EC2, AWS Lambda, and Elastic Beanstalk.
- Select high-performing resilient storage solutions: Understand EBS, S3, and Glacier for different scenarios.
- Design high-performing networks: VPC, CDN solutions like CloudFront, and AWS Global Accelerator.

4. Design Cost-Optimized Architectures (20%)

- Identify cost-effective storage solutions: Leverage S3 Infrequent Access, Glacier, and lifecycle policies.
- Identify cost-effective compute and database services: Use RDS Reserved Instances and EC2 Spot Instances.
- Design cost-optimized network architectures: Efficiently use bandwidth costs and AWS Direct Connect/VPNs.

Hands-on Lab & Assessment: End-to-end deployment using these services

Assessment: Secure, multi-AZ architecture

INFRASTRUCTURE AS CODE (IAC) (1WEEKS)

- Terraform fundamentals: providers, resources, state management
- Implement Terragrunt on top of Terraform.
- Modules, workspaces, and remote state
- CloudFormation basics
- Drift detection & state locking
- Hooks

Hands-on Lab: Provision a VPC + three-tier app on AWS using Terraform

Assessment: Refactor for reusable modules

CONTAINERIZATION WITH DOCKER (1WEEK)

- Docker architecture:
 - Images,
 - containers,
 - Registries,
 - Basics of containerd and runc.
 - Volumes
- Writing optimized Dockerfiles
- Multi-stage builds & caching strategies
- Docker Compose for local development
- Docker Networking

Hands-on Lab: Containerize a web application and push to Docker Hub

Assessment: Build a multi-service stack

KUBERNETES & AWS EKS OR AKS (2WEEKS)

1. Kubernetes Core Concepts

- Architecture: control plane (API server, etcd, controller-manager, scheduler) and worker nodes (kubelet, kube-proxy)
- Objects: Pods, ReplicaSets, Deployments, StatefulSets, DaemonSets
- Config & Secrets: ConfigMaps, Secrets, and environment injection

2. Networking in Kubernetes

- Pod networking models and CNI plugins (Calico, Flannel)
- Services: ClusterIP, NodePort, LoadBalancer
- Ingress controllers and Ingress resources

3. Storage & Configure

- PersistentVolumes, PersistentVolumeClaims, StorageClasses
- Dynamic provisioning vs. static volumes

4. Helm & Packaging

- Helm chart anatomy: Chart.yaml, templates, values
- Chart versioning, dependency management, and repositories

5. Scaling & Resilience

- Horizontal Pod Autoscaler, Cluster Autoscaler
- Managed node group autoscaling
- Self-healing: Pod disruption budgets & Pod priority

6. Security & Compliance

- RBAC: Roles, RoleBindings, ClusterRoles, ClusterRoleBindings
- NetworkPolicy for pod isolation

Hands-on Lab & Assessment

- K8s: Deploy a multi-tier application (frontend, backend, database) with Helm or Kustomize on a local kind/minikube cluster.

Assessment

- Project: Design and deploy a production-grade application on self hosted with:
 - Highly available control plane and worker nodes
 - Secure IAM policies
 - Autoscaling (pod & node)
 - Observability stack (Prometheus + Grafana)
- Deliverables: Terraform/Terraform modules,, a runbook demonstrating failover and scaling tests.

CONFIGURATION MANAGEMENT (1WEEK)

- Ansible fundamentals: playbooks, inventory, modules, Vars Management, Settings.
- Ansible Galaxy.
- Idempotency & as-code best practices
- Hands-on Lab: Configure web servers and deploy code via Ansible dynamic Inventory to ec2 Instances using AWS SSM.
- Assessment: Role-based playbook design

CONTINUOUS INTEGRATION & DELIVERY(CI/CD) (1WEEK)

1. CI/CD Fundamentals

- Concepts: build, test, package, deploy
- Benefits of pipeline-as-code, immutable artifacts, and repeatable runs

2. Jenkins Pipeline

- **Architecture:** master-agent model, executors, plugins and use EKS to install Jenkins.
- **Declarative vs. Scripted Pipelines**
 - Syntax blocks (pipeline, agent, stages, steps)
 - Groovy-based scripted: node, stage, checkout, sh
- **Shared Libraries** for reusable pipeline code
- **Blue Ocean UI** for visualizing pipelines
- **Credentials & Secrets:** using the Credentials store securely
- **Notifications:** email, Slack, Teams integrations

3. Pipeline Best Practices

- Versioning and tagging builds
- Parallelization: splitting tests across agents or runners
- Fail fast vs. full-suite strategies
- Canary & blue-green deploy patterns

Hands-on Labs

- **Jenkins:**
 - Install Jenkins with Docker or on a VM
 - Create a Declarative pipeline that:
 - i. Checks out a sample repo
 - ii. Builds and tests a Java/Python/Node app
 - iii. Publishes a Docker image to a registry
 - iv. Deploys to a dev environment via SSH or Kubernetes plugin

Assessment

- **Project:** Given a microservice repo, implement both:
 - A Jenkins pipeline with rollback capabilities on failure

Deliverables: pipeline definitions, documentation of differences, and comparison of execution times.

OBSERVABILITY & INCIDENT MANAGEMENT (2WEEK)

- Prometheus & Grafana for metrics
- ELK/EFK stack for logs
- Alerting with Alertmanager
- Distributed tracing basics
- Generate custom metrics using python script and also send custom metrics using Push Gateway
- Monitor the status of frontend using black box exporter.

Hands-on Lab: create service monitors and Prometheus rules for alerting to slack of our deployed apps.

Assessment: Implement Prometheus and grafana in EKS cluster.

CHECK OUT OUR FAQ'S

01 Why Enroll in the DevOps Course

The DevOps market surpassed USD 10.4 billion in 2023, with a projected compound annual growth rate (CAGR) of 19.7% from 2024 to 2028. Major companies like ADP, BBC News, eBay, GE, ING, Intuit, PayPal, Splunk, and Uber are utilizing DevOps practices. The average salary for a DevOps Engineer is approximately \$127,231 per year, according to Indeed.com.

02 Who should enroll in this program?

DevOps is open to everyone who had their roots in IT, software development or somewhere along the lines of system administration. This DevOps course is tailored for:

- Individuals aiming to enhance their market credibility and value as experienced DevOps Practitioners
- System Administrators
- Cloud Professionals
- Solutions Architects
- Infrastructure Architects
- Site Reliability Engineers
- Platform Engineers
- Technical Leads
- Software Engineers
- Freshers

03 Are there hands-on exercises in the training?

Yes, our courses include practical labs, real-world projects, and interactive exercises to ensure you gain hands-on experience with the tools and technologies.

CHECK OUT OUR FAQ'S

04 Will I get a certificate after completing the course?

Yes, upon successfully completing the course, you will receive a certificate of completion from StudyBalancer. For certification-focused courses, we also prepare you for the official industry-recognized exams.

05 What if I miss a training session?

If you miss a session, you'll have access to our Learning Management System (LMS), which includes:

- Class recordings
- Presentations & sample code
- Project materials

Additionally, our technical support team is available to help with any questions.

06 What Internet Speed Is Necessary to Attend the Live Sessions?

To ensure a smooth experience during our live sessions, an internet speed of at least 2 Mbps is required.

CAREER ASSISTANCE!

At StudyBalancer, we are dedicated to helping you achieve your career goals with our comprehensive career advancement programs. Whether you're looking to start a new career or advance in your current field, our programs are designed to bridge any skills gaps and provide you with the tools you need to succeed.



Resume
Building

Craft a professional resume that stands out.



LinkedIn
Profiling

Optimize your LinkedIn profile to attract recruiters.



Mock Interview

Practice with industry experts to ace your interviews.



1:1 Mentorship

Receive one-on-one guidance to navigate your career path.



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